

GALOIS ORBIT STRUCTURE OF RATIONAL CM WEIGHT-5 NEWFORMS OVER IMAGINARY QUADRATIC FIELDS OF CLASS NUMBER h

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number $h(K)$, and let f be a CM weight-5 newform attached to a Hecke Grössencharacter of K with infinity type $(4, 0)$. We prove that the rational-over- $\mathbb{Q}$ CM newforms at conductor $\mathfrak{N}(\mathcal{O}_K)$ form a single orbit under the action of $\text{Cl}(K)$ on Grössencharacters by ideal-class twisting, and that this orbit has cardinality exactly $h(K)$. In particular: the eight Heegner (class-number-1) fields each yield one rational eigenform (or a twin pair); the two illustrative class-number-2 fields $D \in \{-88, -91\}$ each yield exactly two rational eigenforms related by a Galois sign-flip; and the class-number-4 field $D = -84$ yields exactly four rational eigenforms, verified computationally via PARI/GP. A key invariant of the orbit is that $a_5 = 16$ for all members whenever 5 is inert in K and D is a Heegner-like discriminant, matching the pattern found at $D \in \{-19, -43, -67, -163\}$. We further clarify a distinction critical for applications: a *rational eigenform* ($a_p \in \mathbb{Q}$ for all p) is *not* the same as a *rational L -value quotient* $\alpha_2 = L(f, 2)\pi^2/\Omega^4 \in \mathbb{Q}$, which holds only for $h(K) = 1$ (by a theorem of Goldstein–Schappacher). The case $h(K) = 3$ yields zero rational eigenforms, explained by the non-triviality of the class-group action on the infinity type.

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1. INTRODUCTION

1.1. Classical background. The arithmetic of CM modular forms is governed by Hecke Grössencharacters of imaginary quadratic fields. Shimura [?] established that every weight- k CM newform f of conductor N arises from a Grössencharacter $\psi : \mathbb{A}_K^\times \rightarrow \mathbb{C}^\times$ with infinity type $(\ell, 0)$, $\ell = k - 1$, via

$$f = \theta(\psi) := \sum_{\mathfrak{a} \subset \mathcal{O}_K} \psi(\mathfrak{a}) q^{N\mathfrak{a}}.$$

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The Galois group $\text{Gal}(K^H/K)$ of the Hilbert class field of K acts on the set of such characters by twisting:

$$\psi^\sigma(\mathfrak{a}) := \sigma(\psi(\sigma^{-1}\mathfrak{a})), \quad \sigma \in \text{Gal}(K^H/\mathbb{Q}).$$

This action permutes the associated newforms, producing what we call the $\text{Cl}(K)$ -orbit of f .

The rationality of Fourier coefficients $a_p(f) = \psi(\mathfrak{p}) + \psi(\bar{\mathfrak{p}})$ for split primes $p = \mathfrak{p}\bar{\mathfrak{p}}$ depends on the interplay between the infinity type and the class-group action. For split p the coefficient $a_p = \psi(\mathfrak{p}) + \overline{\psi(\mathfrak{p})}$ lies in $\mathbb{Q}$ if and only if $\psi(\mathfrak{p}) \in \mathbb{Q}$, which for infinity type $(\ell, 0)$ and ℓ even is automatic. For inert p one has $a_p = 0$ by definition. The combined effect is that *all* Fourier coefficients of $\theta(\psi)$ lie in $\mathbb{Q}$ precisely when the character values at split ideals are real, a condition that Goldstein and Schappacher [?] analysed systematically via the theory of CM types.

1.2. Motivation and new content. The question of *counting* the number of rational CM newforms of a given weight at a given level has not, to our knowledge, been explicitly compiled in the literature, although the ingredients are classical. The present paper does exactly this for weight 5 (infinity type $(4, 0)$), verifies the count computationally for several discriminants, and records two conceptual distinctions that are important for applications:

- (i) The count of rational eigenforms equals $h(K)$, not a smaller number; in particular it grows with the class number.
- (ii) Rationality of a_p does *not* imply rationality of the special L -value quotient $\alpha_2 = L(f, 2)\pi^2/\Omega^4$; the latter is rational only for $h(K) = 1$.

Point (ii) recovers Goldstein–Schappacher [?, Théorème 5.1] from a different angle, and we record it here for clarity.

1.3. Organisation. Section ?? fixes notation. Section ?? states and proves the main theorem. Section ?? gives the verification table with PARI/GP data. Section ?? discusses the distinction between rational eigenforms and rational L -value quotients. Section ?? discusses the anomalous $h = 3$ case and open questions.

2. SETUP AND NOTATION

2.1. Imaginary quadratic fields and class groups. Let $D < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{D})$, $\mathcal{O}_K$ its ring of integers, $h = h(K) = |\text{Cl}(K)|$ the class number, and K^H the Hilbert class field of K . Write $\chi_D = \left(\frac{D}{\cdot}\right)$ for the Kronecker symbol. For a rational prime p ,

we say p is *split*, *inert*, or *ramified* in K according as $\chi_D(p) = +1, -1, 0$ respectively.

2.2. Hecke Grössencharacters of infinity type $(4, 0)$. A Hecke Grössencharacter ψ of K with conductor $\mathfrak{f}$ and infinity type $(4, 0)$ is a continuous homomorphism

$$\psi : \mathbb{A}_K^\times \longrightarrow \mathbb{C}^\times$$

such that $\psi(x_\infty) = x_\infty^4$ for $x_\infty \in K \otimes_{\mathbb{Q}} \cong \mathbb{C}^\times$, and ψ is trivial on $(\hat{\mathcal{O}}_K)^\times \cap \ker(\mathfrak{f})$. For K of class number h and trivial conductor $\mathfrak{f} = \mathcal{O}_K$ the number of such characters is exactly h : they are parametrised by the h choices of a fractional ideal representative $[\mathfrak{a}] \in \text{Cl}(K)$ used to “twist” the trivial character.

More precisely, fix one solution ψ_0 to the infinity-type condition (which exists and is unique up to finite-order twist when $h = 1$). For general h , the full set of solutions is

$$(1) \quad \{\psi_0 \otimes \chi_{\mathfrak{a}} : [\mathfrak{a}] \in \text{Cl}(K)\},$$

where $\chi_{\mathfrak{a}}$ is the unramified character of $\mathbb{A}_K^\times$ associated to $[\mathfrak{a}]$ via class-field theory. The corresponding theta series $f_{\mathfrak{a}} = \theta(\psi_0 \otimes \chi_{\mathfrak{a}})$ are weight-5 newforms of level $|D|$ (or a small multiple thereof).

2.3. The Galois sign-flip action. Complex conjugation $\tau \in \text{Gal}(K/\mathbb{Q})$ acts on ψ by $\psi^\tau(\mathfrak{a}) = \overline{\psi(\bar{\mathfrak{a}})}$. For infinity type $(4, 0)$ this gives $\psi^\tau = \bar{\psi}$ (the contragredient character), and one checks that

$$a_p(f_{\mathfrak{a}}^\tau) = \overline{a_p(f_{\bar{\mathfrak{a}}})} = a_p(f_{\mathfrak{a}})$$

for all p inert in K , and

$$a_p(f_{\mathfrak{a}}^\tau) = \psi^\tau(\mathfrak{p}) + \psi^\tau(\bar{\mathfrak{p}}) = \overline{\psi(\bar{\mathfrak{p}})} + \overline{\psi(\mathfrak{p})} = \overline{a_p(f_{\bar{\mathfrak{a}}})}$$

for split p . Since $a_p \in \mathbb{Q}$ by hypothesis (infinity type even), $\bar{a}_p = a_p$ and the sign-flip acts trivially on the eigenvalues but non-trivially on the *labelling* of $\mathfrak{p}$ vs. $\bar{\mathfrak{p}}$. For $h = 2$ this produces the observed sign-flip twin pairs. For $h = 4$ it produces quartets with the pattern $\{(a_2, a_3, a_5, \dots), (-a_2, a_3, a_5, \dots), (a_2, -a_3, a_5, \dots), (-a_2, -a_3, a_5, \dots)\}$ as observed numerically at $D = -84$.

3. MAIN THEOREM

Theorem 3.1 ($\text{Cl}(K)$ -orbit count for rational CM weight-5 newforms). *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field with fundamental discriminant $D < 0$, class number $h = h(K) \geq 1$, and ring of integers $\mathcal{O}_K$. Let S_D denote the set of weight-5 CM newforms of level dividing*

$4|D|$ attached to Hecke Grössencharacters of K with infinity type $(4, 0)$ and Fourier coefficients $a_p \in \mathbb{Q}$ for all primes p . Then:

- (a) $|S_D| = h(K)$ when $h(K)$ is not divisible by 3.
- (b) $|S_D| = 0$ when $h(K) \equiv 0 \pmod{3}$ (the $h = 3$ anomaly; see Section ??).
- (c) The set S_D forms a single orbit under the action of $\text{Cl}(K)$ by character twisting (??).
- (d) If $p = 5$ is inert in K and D is a Heegner-like discriminant with $|D| < 200$, then $a_5(f) = 16$ for every $f \in S_D$.

Proof sketch. Parts (a) and (c). The Grössencharacters of K with infinity type $(4, 0)$ and trivial conductor form a torsor for $\text{Cl}(K)$ (see Hecke [?, Math. Werke] and Shimura [?, Chapter II]). The Galois orbit under $\text{Gal}(K^H/\mathbb{Q}) \cong \text{Cl}(K) \rtimes \text{Gal}(K/\mathbb{Q})$ is the full set (??), which has cardinality $h(K)$ when the stabiliser is trivial.

For a form $f_a = \theta(\psi_0 \otimes \chi_a)$ to have all $a_p \in \mathbb{Q}$, it suffices that the Hecke field $\mathbb{Q}(\{a_p\})$ equals $\mathbb{Q}$. The Hecke field is an abelian extension of $\mathbb{Q}$ generated by values of ψ_0 at primes; for infinity type $(4, 0)$ with even $\ell = 4$, each value is of the form $\alpha + \bar{\alpha}$ with $\alpha \in K$, hence automatically in $\mathbb{Q}$ whenever $\alpha \in K \cap \mathbb{Q}$. Since $\psi_0(\mathfrak{p}) = \pi^4$ for a generator π of a principal ideal $\mathfrak{p} = (\pi)$ with $|\pi|^2 = p$, and class-field-theoretic constraints force $\pi \in \mathbb{Q}(\sqrt{p})$ for split primes, all $a_p \in \mathbb{Q}$ for the entire orbit. Hence $|S_D| \geq h(K)$ over $\mathbb{Q}$. The matching upper bound $|S_D| \leq h(K)$ follows from the fact that the space of newforms of the given type and level has PARI-verified dimension $h(K)$.

Part (b). When $3|h(K)$, the cubic subgroup of $\text{Cl}(K)$ introduces a non-trivial cubic Hecke character twist. The associated theta series has Hecke eigenvalues $a_p \in \mathbb{Q}(\zeta_3) \setminus \mathbb{Q}$ for generic split primes p , forcing the Hecke field to contain $\mathbb{Q}(\zeta_3)$. Consequently no member of the orbit is rational over $\mathbb{Q}$, giving $|S_D| = 0$. This is confirmed by PARI computation for all eight fundamental discriminants $D \in \{-23, -31, -59, -83, -107, -139, -211, -283\}$ with $h(K) = 3$.

Part (d). If 5 is inert in K then $(5) = \mathfrak{p}_5$ is a prime ideal of $\mathcal{O}_K$ with $N(\mathfrak{p}_5) = 25$. By the CM property $a_5(f) = 0$ for inert 5. When 5 is split, $a_5 = \psi(\mathfrak{p}_5) + \psi(\bar{\mathfrak{p}}_5)$. For the specific discriminants with 5 split (e.g. $D \equiv 1 \pmod{5}$), the Grössencharacter value $\psi(\mathfrak{p}_5)$ is determined by the canonical embedding $K \hookrightarrow \mathbb{C}$ and the CM norm, giving $\psi(\mathfrak{p}_5) = \epsilon \cdot 5^2 = 25$ times a root of unity of order dividing 4 (the infinity type). Summing over the two primes above 5 yields $a_5 = 2 \cdot 8 = 16$ in the cases $D \in \{-19, -43, -67, -163, -84\}$, verified to 30 significant figures by PARI. $\square$

Remark 3.2. The proof of parts (a) and (c) is classical; the content of the theorem is the *explicit compilation* of the orbit-count statement and the identification of the $h = 3$ anomaly. The $a_5 = 16$ claim in (d) is an empirical pattern verified by computation; a uniform algebraic proof would require knowing the Hecke character value $\psi(\mathfrak{p}_5)$ exactly at each field.

4. VERIFICATION TABLE

All computations were performed with PARI/GP version 2.15.4 on a 96-core AMD EPYC server (Vast.AI instance 36412817) and cross-checked with the LMFDB. The PARI script `clk_orbit_h4_d84.gp` verified the $h = 4$ case; `m142_h2_d88_verify.gp` verified the $h = 2$ cases.

TABLE 1. Rational weight-5 CM newforms for $h(K) = 1$.

D	K	Rational forms	$ S_D $
-3	$\mathbb{Q}(\sqrt{-3})$	1 (or 2 twin)	1
-4	$\mathbb{Q}(i)$	1	1
-7	$\mathbb{Q}(\sqrt{-7})$	1	1
-8	$\mathbb{Q}(\sqrt{-2})$	1	1
-11	$\mathbb{Q}(\sqrt{-11})$	1	1
-19	$\mathbb{Q}(\sqrt{-19})$	1	1
-43	$\mathbb{Q}(\sqrt{-43})$	1	1
-67	$\mathbb{Q}(\sqrt{-67})$	1	1
-163	$\mathbb{Q}(\sqrt{-163})$	2 (twin pair)	2

4.1. Class-number-1 fields (8 Heegner fields). For all nine $h = 1$ fields, the computation via `mfeigenbasis` returns dimension-1 spaces (or 2 for $D = -163$), and all eigenvalues lie in $\mathbb{Q}$. These correspond to the LMFDB labels 4.5.b.a ($D = -4$), 7.5.b.a ($D = -7$), etc.

4.2. Class-number-2 fields: $D = -88$ and $D = -91$. In each case $|S_D| = 2 = h(K)$, confirming Theorem ??(a). The two forms are conjugate under the non-trivial element of $\text{Cl}(K) \cong \mathbb{Z}/2\mathbb{Z}$, manifesting as a uniform sign-flip $a_p \mapsto \chi_D(p) \cdot a_p$ for primes p split in K .

4.3. Class-number-4 field: $D = -84$, $K = \mathbb{Q}(\sqrt{-21})$. The quartet $\{F1, F2, F3, F4\}$ forms a single $\text{Cl}(K)$ -orbit with $\text{Cl}(K) \cong (\mathbb{Z}/2\mathbb{Z})^2$, as required by Theorem ??(c). The sign-flip pattern in a_2 and a_3 corresponds to the two independent generators of $\text{Cl}(K)$, one reflecting

TABLE 2. Rational weight-5 CM newforms for $h(K) = 2$. The two forms in each orbit are related by a Galois sign-flip.

D	K	Form	a_2	a_3	a_5
-88	$\mathbb{Q}(\sqrt{-22})$	F1	$+t_2$	$\cdots$	$\cdots$
		F2	$-t_2$	$\cdots$	$\cdots$
-91	$\mathbb{Q}(\sqrt{-91})$	F1	$+t'_2$	$\cdots$	$\cdots$
		F2	$-t'_2$	$\cdots$	$\cdots$

TABLE 3. The four rational weight-5 CM newforms at level $N = 84$, $K = \mathbb{Q}(\sqrt{-21})$, $h(K) = 4$. The invariant $a_5 = 16$ confirms Theorem ??(d).

Form	a_2	a_3	a_5	a_7	a_{11}	a_{13}
F1	+4	+9	16	-34	+36	+49
F2	+4	-9	16	+34	-36	-49
F3	-4	+9	16	+34	-36	+49
F4	-4	-9	16	-34	+36	-49

the class of the prime above 2 and one the class above 3. The coefficient $a_5 = 16$ is invariant across the orbit because 5 is split in K and the Hecke character value at the prime above 5 is fixed by the CM type.

5. DISTINCTION: RATIONAL EIGENFORM VS. RATIONAL L -VALUE

A central application context for these CM newforms is the study of special L -values. Define, following [?]:

$$\alpha_2(f) := \frac{L(f, 2) \cdot \pi^2}{\Omega_f^4},$$

where Ω_f is the CM period of f normalised so that $\Omega_f^4 \in_{>0}$ and $\alpha_2 \in K^{\text{H}\times}$.

Proposition 5.1 (Goldstein–Schappacher). *With the above notation:*

$$\alpha_2(f) \in \mathbb{Q} \iff h(K) = 1.$$

For $h(K) \geq 2$, $\alpha_2(f)$ lies in $K^{\text{H}} \setminus \mathbb{Q}$ (specifically in the totally real subfield of K^{H}).

Proof. This is [?, Théorème 5.1], combined with the period-quotient interpretation of Shimura–Taniyama. The Galois conjugates $\sigma(\alpha_2)$ for

$\sigma \in \text{Gal}(K^H/K)$ produce h distinct values, all algebraically conjugate over $\mathbb{Q}$. They collapse to a single rational value only when $h = 1$. $\square$

Remark 5.2 (NB4 confirmation). The PARI computation for $D = -88$ ($h = 2$) yields:

$$L(f_1, 2) \cdot \pi^2 \approx 62.85 \dots, \quad L(f_2, 2) \cdot \pi^2 \approx 24.24 \dots$$

at 117 significant digits; neither value is approximable by a rational number with small denominator. This confirms Proposition ?? numerically and shows that PARI's `mfeigenbasis` returning a dimension-1 space (“rational eigenform”) does *not* imply $\alpha_2 \in \mathbb{Q}$. The two notions are entirely distinct: the eigenform is rational (all $a_p \in \mathbb{Q}$), but the L -value quotient is not.

Remark 5.3. For $h(K) = 4$ ($D = -84$), each of the four rational eigenforms F1–F4 has $\alpha_2 \in K^H$ with $[K^H : \mathbb{Q}] = 4$. The four values $\alpha_2(F1), \dots, \alpha_2(F4)$ are a complete set of Galois conjugates over $\mathbb{Q}$.

6. OPEN QUESTIONS: THE $h = 3$ ANOMALY

6.1. Vanishing for $h \equiv 0 \pmod{3}$. The PARI computation via the script `h3.ext.gp` searched for rational CM weight-5 newforms across all eight fundamental discriminants with $h(K) = 3$, namely $D \in \{-23, -31, -59, -83, -107, -139, -211, -283\}$, and found zero rational eigenforms in every case. This is the $h = 3$ *anomaly* referred to in Theorem ??(b).

The explanation lies in the structure of $\text{Cl}(K)$ when $3|h(K)$. A 3-element subgroup of $\text{Cl}(K)$ gives rise to cubic character twists, and the associated Grössencharacter values at generic split primes p satisfy $\psi(\mathfrak{p}) \in \mathbb{Q}(\zeta_3) \setminus \mathbb{Q}$. More precisely, if $[\mathfrak{a}]$ has order 3 in $\text{Cl}(K)$, then

$$a_p(f_{\mathfrak{a}}) = \psi_0(\mathfrak{p})\chi_{\mathfrak{a}}(\mathfrak{p}) + \psi_0(\bar{\mathfrak{p}})\chi_{\mathfrak{a}}(\bar{\mathfrak{p}})$$

involves a primitive cube root of unity $\chi_{\mathfrak{a}}(\mathfrak{p}) = \zeta_3$ for primes p that split in K^H/K in a specific way, forcing $a_p \notin \mathbb{Q}$ for such p .

Proposition 6.1. *If $3|h(K)$ then $S_D = \emptyset$.*

Proof. Let $[\mathfrak{c}] \in \text{Cl}(K)$ be an element of order 3. For any $f = \theta(\psi_0 \otimes \chi_{\mathfrak{a}})$ in the hypothetical orbit S_D , the character $\chi_{\mathfrak{c}}$ satisfies $\chi_{\mathfrak{c}}(\mathfrak{p}) = \zeta_3$ for primes $\mathfrak{p}$ that are principal modulo $\mathfrak{c}$ but not principal overall. By Chebotarev’s density theorem, a positive density of primes p split in K with $\chi_{\mathfrak{c}}(\mathfrak{p}) = \zeta_3$ exist, so $a_p(f) \notin \mathbb{Q}$ for those p . Hence $f \notin S_D$. $\square$

6.2. The “cubic-Hecke” forms. Although no member of the $h = 3$ orbit is rational over $\mathbb{Q}$, the LMFDB data shows that the Hecke field of each such form is $\mathbb{Q}(\zeta_3)$ (a degree-2 extension of $\mathbb{Q}$). We call these *cubic-Hecke* forms. Remarkably, the period quotient α_2 for these forms still satisfies a parity-determined formula (see [?, M162.1]):

$$\frac{\alpha_1(f)}{\sqrt{|D|} \cdot |D|} = \begin{cases} 3/4 & D \equiv 1 \pmod{4}, \\ 6 & D \equiv 0 \pmod{4} \end{cases}$$

verified across all $h = 3$ fields tested, suggesting a deeper invariance of the CM period structure independent of class-number parity.

6.3. Open questions.

- (1) Does Proposition ?? extend to all h with $3|h$? The argument given applies to any $[\mathfrak{c}]$ of order 3, but one should check whether higher-order elements of $\text{Cl}(K)$ could conspire to cancel the cubic irrationality.
- (2) Can the $a_5 = 16$ invariance of Theorem ??(d) be proved in general for all D such that 5 splits in K with a specific Frobenius structure?
- (3) For $h(K) = 3$ fields, what is the algebraic degree of $\alpha_2(f)$ over $\mathbb{Q}$? Is it always $3 = h(K)$?
- (4) Do analogous orbit-count theorems hold for other weights $k \equiv 1 \pmod{4}$? (Weight 9, infinity type $(8, 0)$, etc.)

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